

Cat® 3516B

Diesel Generator Sets



Image shown may not reflect actual configuration

Bore – mm (in)	170 (6.69)
Stroke – mm (in)	190 (7.48)
Displacement – L (in ³)	69 (4210.64)
Compression Ratio	14.0:1
Aspiration	TA
Fuel System	EUI
Governor Type	ADEM™ A3

Standby 60 Hz ekW (kVA)	Mission Critical 60 Hz ekW (kVA)	Prime 60 Hz ekW (kVA)	Continuous 60 Hz ekW (kVA)	Emissions Performance
2000 (2500)	2000 (2500)	1825 (2281)	1640 (2050)	Optimized for Low Fuel Consumption and Low Emissions

Features

Cat® Diesel Engine

- Designed and optimized for low emissions or low fuel consumption
- Reliable performance proven in thousands of applications worldwide

Generator Set Package

- Accepts 100% block load in one step and meets NFPA 110 loading requirements
- Conforms to ISO 8528-5 G3 load acceptance requirements
- Reliability verified through torsional vibration, fuel consumption, oil consumption, transient performance, and endurance testing

Alternators

- Superior motor starting capability minimizes need for oversizing generator
- Designed to match performance and output characteristics of Cat diesel engines

Cooling System

- Cooling systems available to operate in ambient temperatures up to 50°C (122°F)
- Tested to ensure proper generator set cooling

EMCP 4 Control Panels

- User-friendly interface and navigation
- Scalable system to meet a wide range of installation requirements
- Expansion modules and site specific programming for specific customer requirements

Warranty

- 24 months/1000-hour warranty for standby and mission critical ratings
- 12 months/unlimited hour warranty for prime and continuous ratings
- Extended service protection is available to provide extended coverage options

Worldwide Product Support

- Cat dealers have over 1,800 dealer branch stores operating in 200 countries
- Your local Cat dealer provides extensive post-sale support, including maintenance and repair agreements

Financing

- Caterpillar offers an array of financial products to help you succeed through financial service excellence
- Options include loans, finance lease, operating lease, working capital, and revolving line of credit
- Contact your local Cat dealer for availability in your region

Standard and Optional Equipment

Engine

Air Cleaner

- ☐ Single element
- ☐ Dual element
- ☐ Heavy duty

Muffler

- ☐ Industrial grade (15 dB)
- ☐ Residential grade (25 dB)
- ☐ Critical grade (35 dB)

Starting

- ☐ Standard batteries
- ☐ Oversized batteries
- ☐ Standard electric starter(s)
- ☐ Dual electric starter(s)
- ☐ Jacket water heater

Alternator

Output voltage

- ☐ 380V ☐ 6900V
- ☐ 400V ☐ 10000V
- ☐ 415V ☐ 10500V
- ☐ 6300V ☐ 11000V
- ☐ 6600V

Temperature Rise (over 40°C ambient)

- ☐ 150°C
- ☐ 125°C/130°C
- ☐ 105°C
- ☐ 80°C

Winding type

- ☐ Random wound
- ☐ Form wound

Excitation

- ☐ Internal excitation (IE)
- ☐ Permanent magnet (PM)

Attachments

- ☐ Anti-condensation heater
- ☐ Stator and bearing temperature monitoring and protection

Power Termination

Type

- ☐ Bus bar
- ☐ Circuit breaker
- ☐ 2000A ☐ 3200A
- ☐ 2500A ☐ 4000A
- ☐ IEC
- ☐ 3-pole
- ☐ Electrically operated

Trip Unit

- ☐ LSI ☐ LSI-G
- ☐ LSIG-P

Control System

Controller

- ☐ EMCP 4.2B
- ☐ EMCP 4.3
- ☐ EMCP 4.4

Attachments

- ☐ Local annunciator module
- ☐ Remote annunciator module
- ☐ Expansion I/O module
- ☐ Remote monitoring software

Charging

- ☐ Battery charger – 10A
- ☐ Battery charger – 20A
- ☐ Battery charger – 35A

Vibration Isolators

- ☐ Rubber
- ☐ Spring
- ☐ Seismic rated

Cat Connect

Connectivity

- ☐ Ethernet
- ☐ Cellular

Extended Service Options

Terms

- ☐ 2 year (prime)
- ☐ 3 year
- ☐ 5 year
- ☐ 10 year

Coverage

- ☐ Silver
- ☐ Gold
- ☐ Platinum
- ☐ Platinum Plus

Ancillary Equipment

- ☐ Automatic transfer switch (ATS)
- ☐ Paralleling switchgear
- ☐ Paralleling controls

Certifications

- ☐ EU Declaration of Conformity
- ☐ EU Declaration of Incorporation
- ☐ Eurasian Conformity (EAC)
- ☐ Telecommunication Lab of China

Note: Some options may not be available on all models. Certifications may not be available with all model configurations. Consult factory for availability.

Package Performance

Low Fuel Consumption (30°C SCAC)

Performance	Standby		Mission Critical		Prime		Continuous	
Frequency	60 Hz		60 Hz		60 Hz		60 Hz	
Gen set power rating with fan	2000 ekW		2000 ekW		1825 ekW		1640 ekW	
Gen set power rating with fan @ 0.8 power factor	2500 kVA		2500 kVA		2281 kVA		2050 kVA	
Emissions	Low Fuel		Low Fuel		Low Fuel		Low Fuel	
Performance number	DM7916-02		EM0648-01		DM7919-02		DM7922-02	
Aftercooler (separate circuit) – °C (°F)	30	(86)	30	(86)	30	(86)	30	(86)
Fuel Consumption								
100% load with fan – L/hr (gal/hr)	613.9	(135.8)	613.9	(135.8)	466.4	(123.2)	421.5	(111.4)
75% load with fan – L/hr (gal/hr)	386.2	(102.0)	386.2	(102.0)	353.6	(93.4)	320.2	(84.6)
50% load with fan – L/hr (gal/hr)	268.2	(70.9)	268.2	(70.9)	248.4	(65.6)	227.6	(60.1)
25% load with fan – L/hr (gal/hr)	158.8	(42.0)	158.8	(42.0)	148.9	(39.3)	138.3	(36.5)
Cooling System								
Radiator air flow restriction (system) – kPa (in. water)	0.12	(0.48)	0.12	(0.48)	0.12	(0.48)	0.12	(0.48)
Radiator air flow – m³/min (cfm)	2348	(82918)	2348	(82918)	2348	(82918)	2348	(82918)
Engine coolant capacity – L (gal)	233.0	(61.6)	233.0	(61.6)	233.0	(61.6)	233.0	(61.6)
Radiator coolant capacity – L (gal)	131.0	(34.6)	131.0	(34.6)	131.0	(34.6)	131.0	(34.6)
Total coolant capacity – L (gal)	364.0	(96.2)	364.0	(96.2)	364.0	(96.2)	364.0	(96.2)
Inlet Air								
Combustion air inlet flow rate – m³/min (cfm)	171.4	(6052.3)	171.4	(6052.3)	164.1	(5794.5)	156.0	(5508.4)
Exhaust System								
Exhaust stack gas temperature – °C (°F)	449.9	(841.8)	449.9	(841.8)	424.7	(796.5)	406.1	(763.0)
Exhaust gas flow rate – m³/min (cfm)	430.6	(15205.1)	430.6	(15205.1)	397.4	(14032.5)	367.2	(12966.0)
Exhaust system backpressure (maximum allowable) – kPa (in. water)	6.7	(27.0)	6.7	(27.0)	6.7	(27.0)	6.7	(27.0)
Heat Rejection								
Heat rejection to jacket water – kW (Btu/min)	733	(41686)	733	(41686)	684	(38898)	634	(36055)
Heat rejection to exhaust (total) – kW (Btu/min)	1881	(106973)	1881	(106973)	1692	(96222)	1516	(86212)
Heat rejection to aftercooler – kW (Btu/min)	555	(31563)	555	(31563)	492	(27980)	426	(24226)
Heat rejection to atmosphere from engine – kW (Btu/min)	136	(7734)	136	(7734)	126	(7165)	119	(6767)
Heat rejection from alternator – kW (Btu/min)	96	(5464)	96	(5464)	86	(4895)	76	(4326)
Emissions* (Nominal)								
NOx mg/Nm³ (g/hp-h)	4032.6	(8.28)	4032.6	(8.28)	4149.6	(8.46)	3795.1	(7.75)
CO mg/Nm³ (g/hp-h)	190.1	(0.39)	190.1	(0.39)	200.9	(0.41)	211.7	(0.43)
HC mg/Nm³ (g/hp-h)	136.7	(0.28)	136.7	(0.28)	126.3	(0.26)	100.1	(0.20)
PM mg/Nm³ (g/hp-h)	44.9	(0.09)	44.9	(0.09)	43.9	(0.09)	44.2	(0.09)
Emissions* (Potential Site Variation)								
NOx mg/Nm³ (g/hp-h)	4839.1	(9.94)	4839.1	(9.94)	4979.5	(10.16)	4554.2	(9.31)
CO mg/Nm³ (g/hp-h)	342.2	(0.70)	342.2	(0.70)	361.6	(0.74)	381.1	(0.78)
HC mg/Nm³ (g/hp-h)	181.8	(0.37)	181.8	(0.37)	168.0	(0.34)	133.1	(0.27)
PM mg/Nm³ (g/hp-h)	62.9	(0.13)	62.9	(0.13)	61.5	(0.13)	61.9	(0.13)

*mg/Nm³ levels are corrected to 5% O₂. Contact your local Cat dealer for further information.

Package Performance

Low Fuel Consumption (60°C SCAC)

Performance	Standby		Mission Critical		Prime		Continuous	
Frequency	60 Hz		60 Hz		60 Hz		60 Hz	
Gen set power rating with fan	2000 ekW		2000 ekW		1825 ekW		1640 ekW	
Gen set power rating with fan @ 0.8 power factor	2500 kVA		2500 kVA		2281 kVA		2050 kVA	
Emissions	Low Fuel		Low Fuel		Low Fuel		Low Fuel	
Performance number	DM7917-02		EM0649-01		DM7920-02		DM7923-01	
Aftercooler (separate circuit) – °C (°F)	60	(140)	60	(140)	60	(140)	60	(140)
Fuel Consumption								
100% load with fan – L/hr (gal/hr)	518.2	(136.9)	518.2	(136.9)	472.7	(124.9)	425.9	(112.5)
75% load with fan – L/hr (gal/hr)	390.1	(103.0)	390.1	(103.0)	356.7	(94.2)	323.8	(85.6)
50% load with fan – L/hr (gal/hr)	272.1	(71.9)	272.1	(71.9)	252.0	(66.6)	231.3	(61.1)
25% load with fan – L/hr (gal/hr)	162.2	(42.8)	162.2	(42.8)	152.1	(40.2)	141.5	(37.4)
Cooling System								
Radiator air flow restriction (system) – kPa (in. water)	0.12	(0.48)	0.12	(0.48)	0.12	(0.48)	0.12	(0.48)
Radiator air flow – m³/min (cfm)	2348	(82918)	2348	(82918)	2348	(82918)	2348	(82918)
Engine coolant capacity – L (gal)	233.0	(61.6)	233.0	(61.6)	233.0	(61.6)	233.0	(61.6)
Radiator coolant capacity – L (gal)	131.0	(34.6)	131.0	(34.6)	131.0	(34.6)	131.0	(34.6)
Total coolant capacity – L (gal)	364.0	(96.2)	364.0	(96.2)	364.0	(96.2)	364.0	(96.2)
Inlet Air								
Combustion air inlet flow rate – m³/min (cfm)	167.2	(5904.0)	167.2	(5904.0)	159.9	(5646.2)	150.9	(5328.3)
Exhaust System								
Exhaust stack gas temperature – °C (°F)	476.4	(889.5)	476.4	(889.5)	455.6	(852.1)	437.9	(820.2)
Exhaust gas flow rate – m³/min (cfm)	438.1	(15469.9)	438.1	(15469.9)	406.6	(14357.3)	374.0	(13206.1)
Exhaust system backpressure (maximum allowable) – kPa (in. water)	6.7	(27.0)	6.7	(27.0)	6.7	(27.0)	6.7	(27.0)
Heat Rejection								
Heat rejection to jacket water – kW (Btu/min)	768	(43676)	768	(43676)	719	(40889)	667	(37931)
Heat rejection to exhaust (total) – kW (Btu/min)	1949	(110840)	1949	(110840)	1778	(101112)	1598	(90876)
Heat rejection to aftercooler – kW (Btu/min)	481	(27355)	481	(27355)	418	(23772)	352	(20018)
Heat rejection to atmosphere from engine – kW (Btu/min)	146	(8303)	146	(8303)	136	(7734)	128	(7279)
Heat rejection from alternator – kW (Btu/min)	96	(5464)	96	(5464)	86	(4895)	76	(4326)
Emissions* (Nominal)								
NOx mg/Nm³ (g/hp-h)	4478.3	(9.27)	4478.3	(9.27)	4495.1	(9.27)	4356.2	(8.99)
CO mg/Nm³ (g/hp-h)	155.9	(0.32)	155.9	(0.32)	145.7	(0.30)	126.6	(0.26)
HC mg/Nm³ (g/hp-h)	134.2	(0.28)	134.2	(0.28)	111.5	(0.23)	97.2	(0.20)
PM mg/Nm³ (g/hp-h)	36.7	(0.08)	36.7	(0.08)	36.6	(0.08)	36.3	(0.07)
Emissions* (Potential Site Variation)								
NOx mg/Nm³ (g/hp-h)	5373.9	(11.12)	5373.9	(11.12)	5394.1	(11.13)	5227.5	(10.79)
CO mg/Nm³ (g/hp-h)	280.6	(0.58)	280.6	(0.58)	262.3	(0.54)	227.9	(0.47)
HC mg/Nm³ (g/hp-h)	178.5	(0.37)	178.5	(0.37)	148.3	(0.31)	129.3	(0.27)
PM mg/Nm³ (g/hp-h)	51.4	(0.11)	51.4	(0.11)	51.2	(0.11)	50.8	(0.10)

*mg/Nm³ levels are corrected to 5% O₂. Contact your local Cat dealer for further information.

Package Performance

Low Fuel Consumption (90°C SCAC)

Performance	Standby		Mission Critical		Prime		Continuous	
Frequency	60 Hz		60 Hz		60 Hz		60 Hz	
Gen set power rating with fan	2000 ekW		2000 ekW		1825 ekW		1640 ekW	
Gen set power rating with fan @ 0.8 power factor	2500 kVA		2500 kVA		2281 kVA		2050 kVA	
Emissions	Low Fuel		Low Fuel		Low Fuel		Low Fuel	
Performance number	DM7918-03		EM0650-01		DM7921-01		DM7924-01	
Aftercooler (separate circuit) – °C (°F)	90	(194)	90	(194)	90	(194)	90	(194)
Fuel Consumption								
100% load with fan – L/hr (gal/hr)	530.8	(140.2)	530.8	(140.2)	479.9	(126.8)	429.7	(113.5)
75% load with fan – L/hr (gal/hr)	392.5	(103.7)	392.5	(103.7)	358.6	(94.7)	324.0	(85.6)
50% load with fan – L/hr (gal/hr)	269.2	(71.1)	269.2	(71.1)	248.2	(65.6)	226.1	(59.7)
25% load with fan – L/hr (gal/hr)	153.8	(40.6)	153.8	(40.6)	143.4	(37.9)	132.5	(35.0)
Cooling System								
Radiator air flow restriction (system) – kPa (in. water)	0.12	(0.48)	0.12	(0.48)	0.12	(0.48)	0.12	(0.48)
Radiator air flow – m³/min (cfm)	2348	(82918)	2348	(82918)	2348	(82918)	2348	(82918)
Engine coolant capacity – L (gal)	233.0	(61.6)	233.0	(61.6)	233.0	(61.6)	233.0	(61.6)
Radiator coolant capacity – L (gal)	131.0	(34.6)	131.0	(34.6)	131.0	(34.6)	131.0	(34.6)
Total coolant capacity – L (gal)	364.0	(96.2)	364.0	(96.2)	364.0	(96.2)	364.0	(96.2)
Inlet Air								
Combustion air inlet flow rate – m³/min (cfm)	164.3	(5801.6)	164.3	(5801.6)	156.7	(5533.2)	147.0	(5190.6)
Exhaust System								
Exhaust stack gas temperature – °C (°F)	514.4	(957.9)	514.4	(957.9)	486.3	(907.3)	464.2	(867.6)
Exhaust gas flow rate – m³/min (cfm)	453.9	(16027.8)	453.9	(16027.8)	416.4	(14703.4)	378.8	(13375.5)
Exhaust system backpressure (maximum allowable) – kPa (in. water)	6.7	(27.0)	6.7	(27.0)	6.7	(27.0)	6.7	(27.0)
Heat Rejection								
Heat rejection to jacket water – kW (Btu/min)	818	(46520)	818	(46520)	762	(43335)	704	(40035)
Heat rejection to exhaust (total) – kW (Btu/min)	2088	(118745)	2088	(118745)	1868	(106230)	1661	(94458)
Heat rejection to aftercooler – kW (Btu/min)	414	(23545)	414	(23545)	351	(19961)	287	(16321)
Heat rejection to atmosphere from engine – kW (Btu/min)	160	(9099)	160	(9099)	148	(8417)	138	(7848)
Heat rejection from alternator – kW (Btu/min)	96	(5464)	96	(5464)	86	(4895)	76	(4326)
Emissions* (Nominal)								
NOx mg/Nm³ (g/hp-h)	4374.5	(9.28)	4374.5	(9.28)	4406.7	(9.25)	4406.3	(9.18)
CO mg/Nm³ (g/hp-h)	232.3	(0.49)	232.3	(0.49)	175.1	(0.37)	150.9	(0.31)
HC mg/Nm³ (g/hp-h)	75.8	(0.16)	75.8	(0.16)	93.5	(0.20)	94.0	(0.20)
PM mg/Nm³ (g/hp-h)	35.1	(0.07)	35.1	(0.07)	34.0	(0.07)	32.6	(0.07)
Emissions* (Potential Site Variation)								
NOx mg/Nm³ (g/hp-h)	5249.4	(11.13)	5249.4	(11.13)	5288.0	(11.10)	5287.6	(11.02)
CO mg/Nm³ (g/hp-h)	418.2	(0.89)	418.2	(0.89)	315.2	(0.66)	271.6	(0.57)
HC mg/Nm³ (g/hp-h)	100.8	(0.21)	100.8	(0.21)	124.4	(0.26)	125.0	(0.26)
PM mg/Nm³ (g/hp-h)	49.1	(0.10)	49.1	(0.10)	47.6	(0.10)	45.6	(0.10)

*mg/Nm³ levels are corrected to 5% O₂. Contact your local Cat dealer for further information.

Package Performance

Low Emissions (30°C SCAC)

Performance	Standby		Mission Critical		Prime		Continuous	
Frequency	60 Hz		60 Hz		60 Hz		60 Hz	
Gen set power rating with fan	2000 ekW		2000 ekW		1825 ekW		1640 ekW	
Gen set power rating with fan @ 0.8 power factor	2500 kVA		2500 kVA		2281 kVA		2050 kVA	
Emissions	Low Emissions		Low Emissions		Low Emissions		Low Emissions	
Performance number	DM7925-04		EM0652-01		DM7928-02		DM7931-02	
Aftercooler (separate circuit) – °C (°F)	30	(86)	30	(86)	30	(86)	30	(86)
Fuel Consumption								
100% load with fan – L/hr (gal/hr)	558.3	(147.5)	558.3	(147.5)	503.9	(133.1)	450.7	(119.1)
75% load with fan – L/hr (gal/hr)	411.6	(108.7)	411.6	(108.7)	376.2	(99.4)	339.7	(98.7)
50% load with fan – L/hr (gal/hr)	282.4	(74.6)	282.4	(74.6)	260.4	(68.8)	237.3	(62.7)
25% load with fan – L/hr (gal/hr)	162.1	(42.8)	162.1	(42.8)	151.5	(40.0)	140.4	(37.1)
Cooling System								
Radiator air flow restriction (system) – kPa (in. water)	0.12	(0.48)	0.12	(0.48)	0.12	(0.48)	0.12	(0.48)
Radiator air flow – m³/min (cfm)	2348	(82918)	2348	(82918)	2348	(82918)	2348	(82918)
Engine coolant capacity – L (gal)	233.0	(61.6)	233.0	(61.6)	233.0	(61.6)	233.0	(61.6)
Radiator coolant capacity – L (gal)	131.0	(34.6)	131.0	(34.6)	131.0	(34.6)	131.0	(34.6)
Total coolant capacity – L (gal)	364.0	(96.2)	364.0	(96.2)	364.0	(96.2)	364.0	(96.2)
Inlet Air								
Combustion air inlet flow rate – m³/min (cfm)	179.6	(6341.8)	179.6	(6341.8)	174.4	(6158.2)	166.4	(5875.7)
Exhaust System								
Exhaust stack gas temperature – °C (°F)	504.3	(939.8)	504.3	(939.8)	465.0	(869.0)	434.5	(814.1)
Exhaust gas flow rate – m³/min (cfm)	486.4	(17175.5)	486.4	(17175.5)	447.3	(15794.5)	408.3	(14417.2)
Exhaust system backpressure (maximum allowable) – kPa (in. water)	6.7	(27.0)	6.7	(27.0)	6.7	(27.0)	6.7	(27.0)
Heat Rejection								
Heat rejection to jacket water – kW (Btu/min)	779	(44302)	779	(44302)	723	(41116)	667	(37931)
Heat rejection to exhaust (total) – kW (Btu/min)	2211	(125741)	2211	(125741)	1960	(111463)	1728	(98268)
Heat rejection to aftercooler – kW (Btu/min)	635	(36113)	635	(36113)	567	(32245)	494	(28093)
Heat rejection to atmosphere from engine – kW (Btu/min)	153	(8701)	153	(8701)	139	(7905)	128	(7279)
Heat rejection from alternator – kW (Btu/min)	96	(5464)	96	(5464)	86	(4895)	76	(4326)
Emissions* (Nominal)								
NOx mg/Nm³ (g/hp-h)	2101.4	(4.68)	2101.4	(4.68)	2112.6	(4.65)	2095.2	(4.57)
CO mg/Nm³ (g/hp-h)	207.9	(0.46)	207.9	(0.46)	236.8	(0.52)	233.5	(0.51)
HC mg/Nm³ (g/hp-h)	115.3	(0.26)	115.3	(0.26)	118.5	(0.26)	121.1	(0.26)
PM mg/Nm³ (g/hp-h)	55.9	(0.12)	55.9	(0.12)	56.8	(0.13)	59.7	(0.13)
Emissions* (Potential Site Variation)								
NOx mg/Nm³ (g/hp-h)	2521.7	(5.62)	2521.7	(5.62)	2535.1	(5.58)	2514.2	(5.49)
CO mg/Nm³ (g/hp-h)	374.2	(0.83)	374.2	(0.83)	426.2	(0.94)	2420.3	(0.92)
HC mg/Nm³ (g/hp-h)	153.3	(0.34)	153.3	(0.34)	157.6	(0.35)	161.1	(0.35)
PM mg/Nm³ (g/hp-h)	78.3	(0.17)	78.3	(0.17)	79.5	(0.18)	83.6	(0.18)

*mg/Nm³ levels are corrected to 5% O₂. Contact your local Cat dealer for further information.

Package Performance

Low Emissions (60°C SCAC)

Performance	Standby		Mission Critical		Prime		Continuous	
Frequency	60 Hz		60 Hz		60 Hz		60 Hz	
Gen set power rating with fan	2000 ekW		2000 ekW		1825 ekW		1640 ekW	
Gen set power rating with fan @ 0.8 power factor	2500 kVA		2500 kVA		2281 kVA		2050 kVA	
Emissions	Low Emissions		Low Emissions		Low Emissions		Low Emissions	
Performance number	DM7926-03		EM0653-01		DM7929-01		DM7932-01	
Aftercooler (separate circuit) – °C (°F)	60	(140)	60	(140)	60	(140)	60	(140)
Fuel Consumption								
100% load with fan – L/hr (gal/hr)	558.5	(147.5)	558.5	(147.5)	505.5	(133.5)	451.5	(119.3)
75% load with fan – L/hr (gal/hr)	411.7	(108.8)	411.7	(108.8)	375.7	(99.2)	340.0	(89.8)
50% load with fan – L/hr (gal/hr)	283.6	(74.9)	283.6	(74.9)	262.0	(69.2)	239.2	(63.2)
25% load with fan – L/hr (gal/hr)	164.7	(43.5)	164.7	(43.5)	154.1	(40.7)	142.9	(37.7)
Cooling System								
Radiator air flow restriction (system) – kPa (in. water)	0.12	(0.48)	0.12	(0.48)	0.12	(0.48)	0.12	(0.48)
Radiator air flow – m³/min (cfm)	2348	(82918)	2348	(82918)	2348	(82918)	2348	(82918)
Engine coolant capacity – L (gal)	233.0	(61.6)	233.0	(61.6)	233.0	(61.6)	233.0	(61.6)
Radiator coolant capacity – L (gal)	131.0	(34.6)	131.0	(34.6)	131.0	(34.6)	131.0	(34.6)
Total coolant capacity – L (gal)	364.0	(96.2)	364.0	(96.2)	364.0	(96.2)	364.0	(96.2)
Inlet Air								
Combustion air inlet flow rate – m³/min (cfm)	175.6	(6200.6)	175.6	(6200.6)	169.7	(5992.2)	161.2	(5692.1)
Exhaust System								
Exhaust stack gas temperature – °C (°F)	522.9	(973.2)	522.9	(973.2)	487.2	(909.0)	457.3	(855.1)
Exhaust gas flow rate – m³/min (cfm)	488.8	(17260.2)	488.8	(17260.2)	450.3	(15900.4)	410.2	(14484.3)
Exhaust system backpressure (maximum allowable) – kPa (in. water)	6.7	(27.0)	6.7	(27.0)	6.7	(27.0)	6.7	(27.0)
Heat Rejection								
Heat rejection to jacket water – kW (Btu/min)	811	(46121)	811	(46121)	754	(42879)	696	(39580)
Heat rejection to exhaust (total) – kW (Btu/min)	2255	(128243)	2255	(128243)	2007	(114136)	1774	(100884)
Heat rejection to aftercooler – kW (Btu/min)	551	(31336)	551	(31336)	487	(27694)	418	(23771)
Heat rejection to atmosphere from engine – kW (Btu/min)	162	(9213)	162	(9213)	147	(8361)	135	(7677)
Heat rejection from alternator – kW (Btu/min)	96	(5464)	96	(5464)	86	(4895)	76	(4326)
Emissions* (Nominal)								
NOx mg/Nm³ (g/hp-h)	2284.7	(5.09)	2284.7	(5.09)	2567.5	(5.67)	2576.8	(5.64)
CO mg/Nm³ (g/hp-h)	292.0	(0.65)	292.0	(0.65)	160.0	(0.35)	163.0	(0.36)
HC mg/Nm³ (g/hp-h)	38.3	(0.09)	38.3	(0.09)	105.9	(0.23)	117.4	(0.26)
PM mg/Nm³ (g/hp-h)	57.2	(0.13)	57.2	(0.13)	49.5	(0.11)	46.6	(0.10)
Emissions* (Potential Site Variation)								
NOx mg/Nm³ (g/hp-h)	2741.6	(6.11)	2741.6	(6.11)	3081.0	(6.80)	3092.2	(6.76)
CO mg/Nm³ (g/hp-h)	525.6	(1.17)	525.6	(1.17)	288.0	(0.64)	293.4	(0.64)
HC mg/Nm³ (g/hp-h)	50.9	(0.11)	50.9	(0.11)	140.8	(0.31)	156.1	(0.34)
PM mg/Nm³ (g/hp-h)	80.1	(0.18)	80.1	(0.18)	69.3	(0.15)	65.2	(0.14)

*mg/Nm³ levels are corrected to 5% O₂. Contact your local Cat dealer for further information.

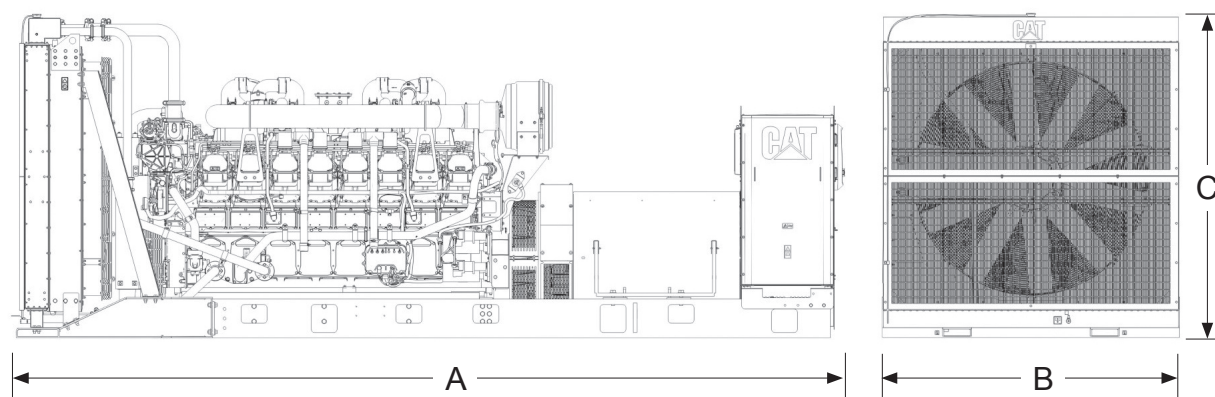
Package Performance

Low Emissions (90°C SCAC)

Performance	Standby		Mission Critical		Prime		Continuous	
Frequency	60 Hz		60 Hz		60 Hz		60 Hz	
Gen set power rating with fan	2000 ekW		2000 ekW		1825 ekW		1640 ekW	
Gen set power rating with fan @ 0.8 power factor	2500 kVA		2500 kVA		2281 kVA		2050 kVA	
Emissions	Low Emissions		Low Emissions		Low Emissions		Low Emissions	
Performance number	DM7927-04		EM0654-01		DM7930-01		DM7933-01	
Aftercooler (separate circuit) – °C (°F)	90	(194)	90	(194)	90	(194)	90	(194)
Fuel Consumption								
100% load with fan – L/hr (gal/hr)	544.9	(144.0)	544.9	(144.0)	496.6	(68.7)	446.0	(117.8)
75% load with fan – L/hr (gal/hr)	408.3	(107.9)	408.3	(107.9)	373.6	(98.7)	338.7	(89.5)
50% load with fan – L/hr (gal/hr)	282.2	(74.6)	282.2	(74.6)	260.2	(68.7)	236.5	(62.5)
25% load with fan – L/hr (gal/hr)	156.5	(41.3)	156.5	(41.3)	144.8	(38.2)	132.3	(35.0)
Cooling System								
Radiator air flow restriction (system) – kPa (in. water)	0.12	(0.48)	0.12	(0.48)	0.12	(0.48)	0.12	(0.48)
Radiator air flow – m³/min (cfm)	2348	(82918)	2348	(82918)	2348	(82918)	2348	(82918)
Engine coolant capacity – L (gal)	233.0	(61.6)	233.0	(61.6)	233.0	(61.6)	233.0	(61.6)
Radiator coolant capacity – L (gal)	131.0	(34.6)	131.0	(34.6)	131.0	(34.6)	131.0	(34.6)
Total coolant capacity – L (gal)	364.0	(96.2)	364.0	(96.2)	364.0	(96.2)	364.0	(96.2)
Inlet Air								
Combustion air inlet flow rate – m³/min (cfm)	168.1	(5935.8)	168.1	(5935.8)	161.6	(5706.2)	153.1	(5406.0)
Exhaust System								
Exhaust stack gas temperature – °C (°F)	525.7	(978.3)	525.7	(978.3)	498.9	(930.0)	474.5	(886.1)
Exhaust gas flow rate – m³/min (cfm)	470.5	(16614.0)	470.5	(16614.0)	436.7	(15420.2)	399.9	(14120.6)
Exhaust system backpressure (maximum allowable) – kPa (in. water)	6.7	(27.0)	6.7	(27.0)	6.7	(27.0)	6.7	(27.0)
Heat Rejection								
Heat rejection to jacket water – kW (Btu/min)	834	(47429)	834	(47429)	779	(44302)	723	(41116)
Heat rejection to exhaust (total) – kW (Btu/min)	2186	(124318)	2186	(124318)	1973	(112203)	1769	(100600)
Heat rejection to aftercooler – kW (Btu/min)	449	(25535)	449	(25535)	392	(22293)	330	(18766)
Heat rejection to atmosphere from engine – kW (Btu/min)	168	(9554)	168	(9554)	155	(8816)	144	(8189)
Heat rejection from alternator – kW (Btu/min)	96	(5464)	96	(5464)	86	(4895)	76	(4326)
Emissions* (Nominal)								
NOx mg/Nm³ (g/hp-h)	2544.0	(5.54)	2544.0	(5.54)	3596.2	(7.07)	3609.2	(7.09)
CO mg/Nm³ (g/hp-h)	258.9	(0.56)	258.9	(0.56)	171.0	(0.57)	156.7	(0.34)
HC mg/Nm³ (g/hp-h)	47.7	(0.10)	47.7	(0.10)	62.8	(0.10)	103.3	(0.22)
PM mg/Nm³ (g/hp-h)	51.1	(0.11)	51.1	(0.11)	41.4	(0.11)	41.8	(0.09)
Emissions* (Potential Site Variation)								
NOx mg/Nm³ (g/hp-h)	3052.8	(6.65)	3052.8	(6.65)	4315.4	(8.48)	4331.0	(8.51)
CO mg/Nm³ (g/hp-h)	466.0	(1.02)	466.0	(1.02)	307.8	(1.03)	282.1	(0.61)
HC mg/Nm³ (g/hp-h)	63.4	(0.14)	63.4	(0.14)	83.5	(0.14)	137.4	(0.30)
PM mg/Nm³ (g/hp-h)	71.5	(0.16)	71.5	(0.16)	58.0	(0.16)	58.5	(0.13)

*mg/Nm³ levels are corrected to 5% O₂. Contact your local Cat dealer for further information.

Weights and Dimensions



Dim "A" mm (in)	Dim "B" mm (in)	Dim "C" mm (in)	Dry Weight kg (lb)
5957 (234.5)	2286 (90.0)	2494 (98.2)	13 750 (30,310)

Note: For reference only. Do not use for installation design. Contact your local Cat dealer for precise weights and dimensions.

Ratings Definitions

Standby

Output available with varying load for the duration of the interruption of the normal source power. Average power output is 70% of the standby power rating. Typical operation is 200 hours per year, with maximum expected usage of 500 hours per year.

Mission Critical

Output available with varying load for the duration of the interruption of the normal source power. Average power output is 85% of the mission critical power rating. Typical peak demand up to 100% of rated power for up to 5% of the operating time. Typical operation is 200 hours per year, with maximum expected usage of 500 hours per year.

Prime

Output available with varying load for an unlimited time. Average power output is 70% of the prime power rating. Typical peak demand is 100% of prime rated kW with 10% overload capability for emergency use for a maximum of 1 hour in 12. Overload operation cannot exceed 25 hours per year.

Continuous

Output available with non-varying load for an unlimited time. Average power output is 70-100% of the continuous power rating. Typical peak demand is 100% of continuous rated kW for 100% of the operating hours.

Applicable Codes and Standards

AS 1359, CSA C22.2 No. 100-04, UL 142, UL 489, UL 869, UL 2200, IBC, IEC 60034-1, ISO 3046, ISO 8528, NEMA MG1-22, NEMA MG1-33, 2014/35/EU, 2006/42/EC, 2014/30/EU and facilitates compliance to NFPA 37, NFPA 70, NFPA 99, NFPA 110.

Note: Codes may not be available in all model configurations. Please consult your local Cat dealer for availability.

Data Center Applications

- ISO 8528-1 Data Center Power (DCP) compliant per DCP application of Cat diesel generator set prime power rating.
- All ratings Tier III/Tier IV compliant per Uptime Institute requirements.
- All ratings ANSI/TIA-942 compliant for Rated-1 through Rated-4 data centers.

Fuel Rates

Fuel rates are based on fuel oil of 35° API [16°C (60°F)] gravity having an LHV of 42,780 kJ/kg (18,390 Btu/lb) when used at 29°C (85°F) and weighing 838.9 g/liter (7.001 lbs/U.S. gal.)

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The International System of Units (SI) is used in this publication.



The 3516B generator set has been developed for a wide range of applications, from emergency standby installations such as healthcare and datacenters to continuously powering remote installations. The packages can be optimized for performance that matters to you with either low emissions or low fuel consumption versions available. Backed by the worldwide network of Cat dealers ready to support your operation with technical support, service, parts, and warranty, Cat generator sets will provide the reliability and durability you expect.

Specifications

Generator Set Specifications	
Minimum Rating	1600 ekW (1750 kVA)
Maximum Rating	2000 ekW (2500 kVA)
Voltage	400 to 11,000 Volts
Frequency	50 Hz
Speed	1500 RPM
Duty Cycle	Standby, Prime, Continuous

Generator Set Configurations	
Emissions/Fuel Strategy	Low Fuel Consumption, Low Emissions

Engine Specifications		
Engine Model	3516B TA, V-16, 4-Stroke Water-Cooled Diesel	
Bore	170 mm	6.69 in
Displacement (Std)	69 L	4210.64 in3
Displacement (HD)	78.08 L	4764.73 in3
Stroke (Std)	190 mm	7.48 in
Stroke (HD)	215 mm	8.46 in
Compression Ratio	14.0:1 Std 15.5:1 HD	
Aspiration	TA	
Governor Type	Adem™3	
Fuel System	Electronic unit injection	
Exhaust Flange Size	203.2 mm (8.0 in)	
Air Inlet	Single element canister style with service indicator	

Benefits and Features

Cat Generator Set Package

Cat generator set packages have been fully prototype tested, and certified torsional vibration analysis reports are available. The packages are designed to accept 100% load in one step, meet the NFPA 110 requirement for loading, and conform to the ISO 8528-5 steady state and transient response requirements

Cat Diesel Engine

The four cycle Cat diesel engine combines consistent performance with excellent fuel economy and transient response that meets or exceeds ISO 8528-5. The engines have been designed and built for a wide range of applications and can be optimized for low fuel consumption or low emissions. The engines feature a reliable, rugged, and durable design that has been field proven in thousands of applications worldwide from emergency standby installations to continuously operating power plants.

Cooling System

The cooling system has been designed to operate in standard ambient temperatures up to 50°C (122°F), with optional high ambient radiators available. The factory installed cooling system has been designed and tested to ensure proper generator set cooling, and includes the radiator, fan, belts, and all guarding installed as standard. Contact your Cat Dealer for specific ambient and altitude capabilities.

Generators

The generators used on Cat packages have been designed and tested to work with the Cat engine. The generators are built with robust Class H insulation and provide industry leading motor starting capability. Random wound generators provide good generator performance in a majority of applications and form wound is available for harsh mechanical and electrical environments.

Cat EMCP Control Panel

The EMCP controller features the reliability and durability you have come to expect from your Cat equipment. EMCP4 is a scalable control platform designed to ensure reliable generator set operation, providing extensive information about power output and engine operation. EMCP4 systems can be further customized to meet your needs through programming and expansion modules.

World Wide Product Support

Cat Dealers provide extensive post sale support including maintenance and repair agreements. Cat dealers have over 1,800 dealer branch stores operating in 200 countries. The Caterpillar S•O•SSM program cost effectively detects internal engine component condition, even the presence of unwanted fluids and combustion by-products.

Optional Equipment

Engine Options

- Air Cleaner: ☐ Single element air cleaners ☐ Dual element air cleaners ☐ Heavy duty air cleaners
- Muffler: ☐ Industrial Grade
- Batteries: ☐ Standard (1400 CCA) ☐ Heavy Duty (2800 CCA)
- Starting Motors: ☐ Standard ☐ Heavy Duty
- Battery Charger: ☐ 10A ☐ 20A ☐ 35A
- Starting Aids: ☐ 220V 50 Hz Jacket Water Heater
- Vibration Isolator: ☐ Rubber anti-vibration mounts (90% efficient) ☐ Spring type anti-vibration mounts (95% efficient)

Control System

- Controller: ☐ EMCP 4.2 ☐ EMCP 4.3 ☐ EMCP 4.4
- Local annunciator module: ☐ NFPA 110 ☐ Custom

- Remote annunciator module: ☐ NFPA 110 ☐ Custom
- Additional Options: ☐ Expansion I/O module ☐ Remote monitoring software

Alternators

- Temperature Rise (over 40°C ambient): ☐ 150°C ☐ 125°C ☐ 105°C ☐ 80°C
- Winding: ☐ Random ☐ Form
- Excitation: ☐ Permanent Magnet (PM) excited ☐ Internally Excited (IE)
- Alternator attachments: ☐ Anti-condensation heaters ☐ Generator stator and bearing temperature monitoring & protection

Power Termination

- Power Termination: ☐ Bus Bar ☐ Circuit Breaker
- Circuit Breaker rating: ☐ 2000A ☐ 2500A ☐ 3200A ☐ 4000A
- Circuit breaker trip unit: ☐ LSI ☐ LSIG ☐ LSIG-P
- Circuit breaker operation: ☐ Electrical

Extended Service Contract (ESC)

- Extended Service Contract (ESC): ☐ 2 Year ☐ 3 Year ☐ 5 Year

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3516B

1600 ekW/ 2000 kVA/ 50 Hz/ 1500 rpm/ 11000 V/ 0.8 Power Factor

Rating Type: CONTINUOUS

Fuel Strategy: LOW FUEL CONSUMPTION

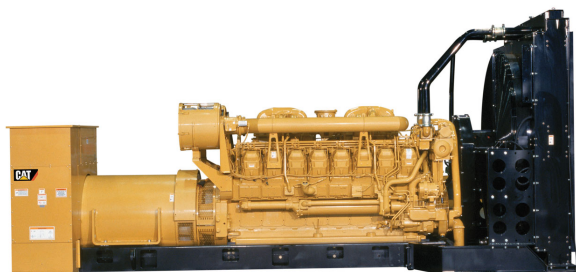


Image shown may not reflect actual configuration

3516B

1600 ekW/ 2000 kVA
50 Hz/ 1500 rpm/ 11000 V

Metric

English

Package Performance

Genset Power Rating with Fan @ 0.8 Power Factor	1600 ekW	
Genset Power Rating	2000 kVA	
Aftercooler (Separate Circuit)	90.0 ° C	194.0 ° F

Fuel Consumption

100% Load with Fan	409.1 L/hr	108.1 gal/hr
75% Load with Fan	305.3 L/hr	80.6 gal/hr
50% Load with Fan	213.8 L/hr	56.5 gal/hr
25% Load with Fan	122.9 L/hr	32.5 gal/hr

Cooling System¹

Engine Coolant Capacity	233.0 L	61.6 gal
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Inlet Air

Combustion Air Inlet Flow Rate	129.7 m ³ /min	4579.9 cfm
Max. Allowable Combustion Air Inlet Temp	99 ° C	210 ° F

Exhaust System

Exhaust Stack Gas Temperature	508.8 ° C	947.8 ° F
Exhaust Gas Flow Rate	356.7 m ³ /min	12595.6 cfm
Exhaust System Backpressure (Maximum Allowable)	6.7 kPa	27.0 in. water

3516B

1600 ekW/ 2000 kVA/ 50 Hz/ 1500 rpm/ 11000 V/ 0.8 Power Factor

Rating Type: CONTINUOUS

Fuel Strategy: LOW FUEL CONSUMPTION

Heat Rejection

Heat Rejection to Jacket Water	640 kW	36397 Btu/min
Heat Rejection to Exhaust (Total)	1646 kW	93608 Btu/min
Heat Rejection to Aftercooler	239 kW	13592 Btu/min
Heat Rejection to Atmosphere from Engine	147 kW	8360 Btu/min
Heat Rejection to Atmosphere from Generator	68 kW	3890 Btu/min

Alternator²

Motor Starting Capability @ 30% Voltage Dip	4196 skVA
Current	105 amps
Frame Size	2770
Excitation	PM
Temperature Rise	105 ° C

Emissions (Nominal)³

NOx	3446.1 mg/Nm ³	7.1 g/hp-hr
CO	163.3 mg/Nm ³	0.3 g/hp-hr
HC	70.7 mg/Nm ³	0.1 g/hp-hr
PM	14.3 mg/Nm ³	0.0 g/hp-hr

DEFINITIONS AND CONDITIONS

1. For ambient and altitude capabilities consult your Cat dealer. Air flow restriction (system) is added to existing restriction from factory.
2. UL 2200 Listed packages may have oversized generators with a different temperature rise and motor starting characteristics. Generator temperature rise is based on a 40° C ambient per NEMA MG1-32.
3. Emissions data measurement procedures are consistent with those described in EPA CFR 40 Part 89, Subpart D & E and ISO8178-1 for measuring HC, CO, PM, NOx. Data shown is based on steady state operating conditions of 77° F, 28.42 in HG and number 2 diesel fuel with 35° API and LHV of 18,390 btu/lb. The nominal emissions data shown is subject to instrumentation, measurement, facility and engine to engine variations. Emissions data is based on 100% load and thus cannot be used to compare to EPA regulations which use values based on a weighted cycle.

3516B

1600 ekW/ 2000 kVA/ 50 Hz/ 1500 rpm/ 11000 V/ 0.8 Power Factor

Rating Type: CONTINUOUS

Fuel Strategy: LOW FUEL CONSUMPTION

Applicable Codes and Standards:

AS1359, CSA C22.2 No100-04, UL142, UL489, UL869, UL2200,
NFPA37, NFPA70, NFPA99, NFPA110, IBC, IEC60034-1, ISO3046, ISO8528,
NEMA MG1-22, NEMA MG1-33, 2006/95/EC, 2006/42/EC, 2004/108/EC.

Note: Codes may not be available in all model configurations. Please consult your local Cat Dealer representative for availability.

CONTINUOUS: Output available with non-varying load for an unlimited time. Average power output is 70-100% of the continuous power rating. Typical peak demand is 100% of continuous rated ekW for 100% of operating hours.

Ratings are based on SAE J1349 standard conditions. These ratings also apply at ISO3046 standard conditions

Fuel Rates are based on fuel oil of 35° API [16° C (60° F)] gravity having an LHV of 42 780 kJ/kg (18,390 Btu/lb) when used at 29° C (85° F) and weighing 838.9 g/liter (7.001 lbs/U.S. gal.). Additional ratings may be available for specific customer requirements, contact your Cat representative for details. For information regarding Low Sulfur fuel and Biodiesel capability, please consult your Cat dealer.

www.Cat-ElectricPower.com

Performance No.: DM7972-01

Feature Code: 516DR8A

Generator Arrangement: 2524246

Date: 08/30/2016

Source Country: CHINA

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